



Batterjee Medical College
For Science and Technology

Regulatory systems

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بِسْمِ اللَّهِ الرَّحْمَنِ
الرَّحِيمِ



وَاتَّقُوا اللَّهَ^{صلى} ^{قله} وَاعْلَمُوا أَنَّ اللَّهَ^{صلى} قَدِيرٌ

Learning outcomes:

By the end of the lectures, the student will be able to:

- **Compare the two types of regulatory systems in the body**
- **Identify the structure and functional unit of the nervous system**
- **List the endocrine glands in the body and the hormones secreted by each gland**



Regulatory systems

Regulatory systems

	<u>The endocrine system</u>	<u>The nervous system</u>
Structural unit		
Functional unit		
Onset		
Duration		
Example		



Nervous system

Nervous system

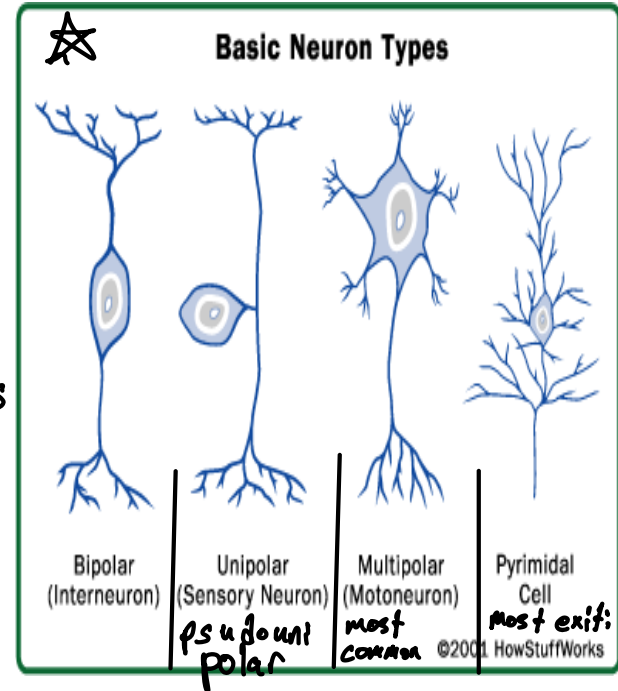
? *The nervous system* is comprised of billions of cells, which are of two major cell types:

1. **Neurons** (the basic structural unit of nervous tissue, specialized for transmission of APs).
2. **Glial cells** (Neuroglia, supporting cells, the Greek of glue).

↳ is the structure unit of ns

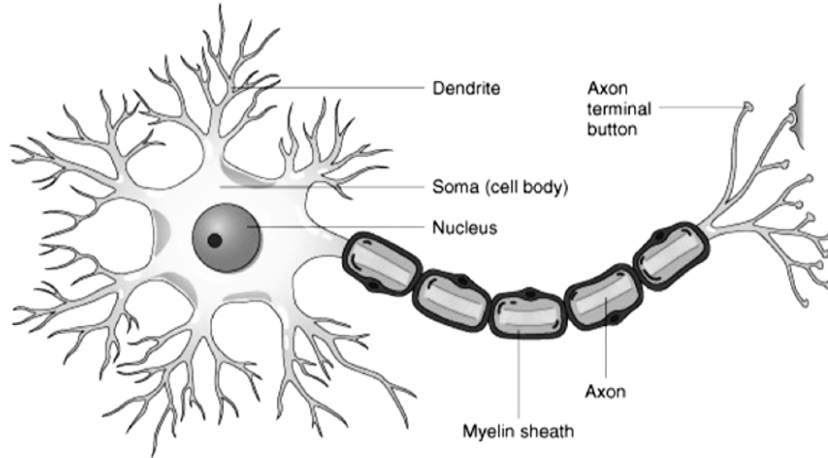
? Neurons in the mammalian nervous system come in many different shapes and sizes.

? However, most have the same parts as the typical spinal motor neuron (multipolar). It is composed of three major parts:



Nervous system

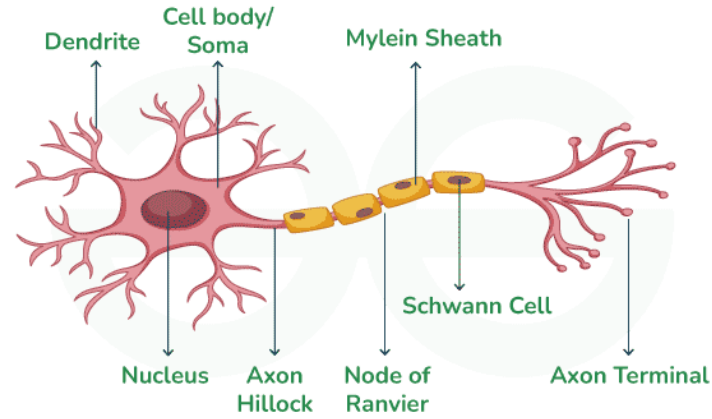
- a. **The Soma:** is the main cell body of the neuron. It acts as a processing center.
- b. **The Dendrites:** are antenna-like processes that project out from the soma and increase the cell surface area for reception of signals from other neurons.



- c. **Axon (Nerve Fiber)** a single more elongated process which extends for a distance.

Nervous system

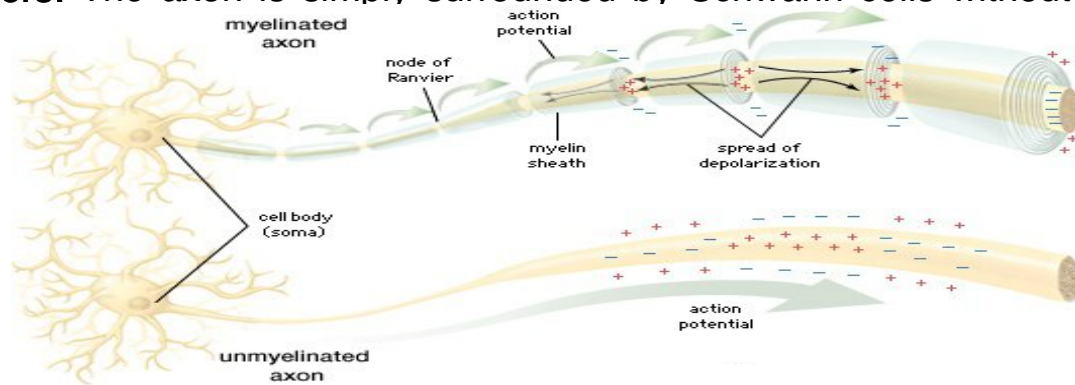
- The axon originates from a somewhat thickened area of the cell body, the **axon hillock**.
- The first portion of the axon is called the **initial segment**.
- The axon is composed of axis cylinder surrounded with the plasma membrane, which is a continuation of the cell membrane
- The axon ends in number of synaptic Knobs, which are also called **terminal buttons**.
- These knobs contain granules or vesicles in which the synaptic transmitters secreted by the nerves are stored.
- Impulses are directed away from the cell body by its axon.



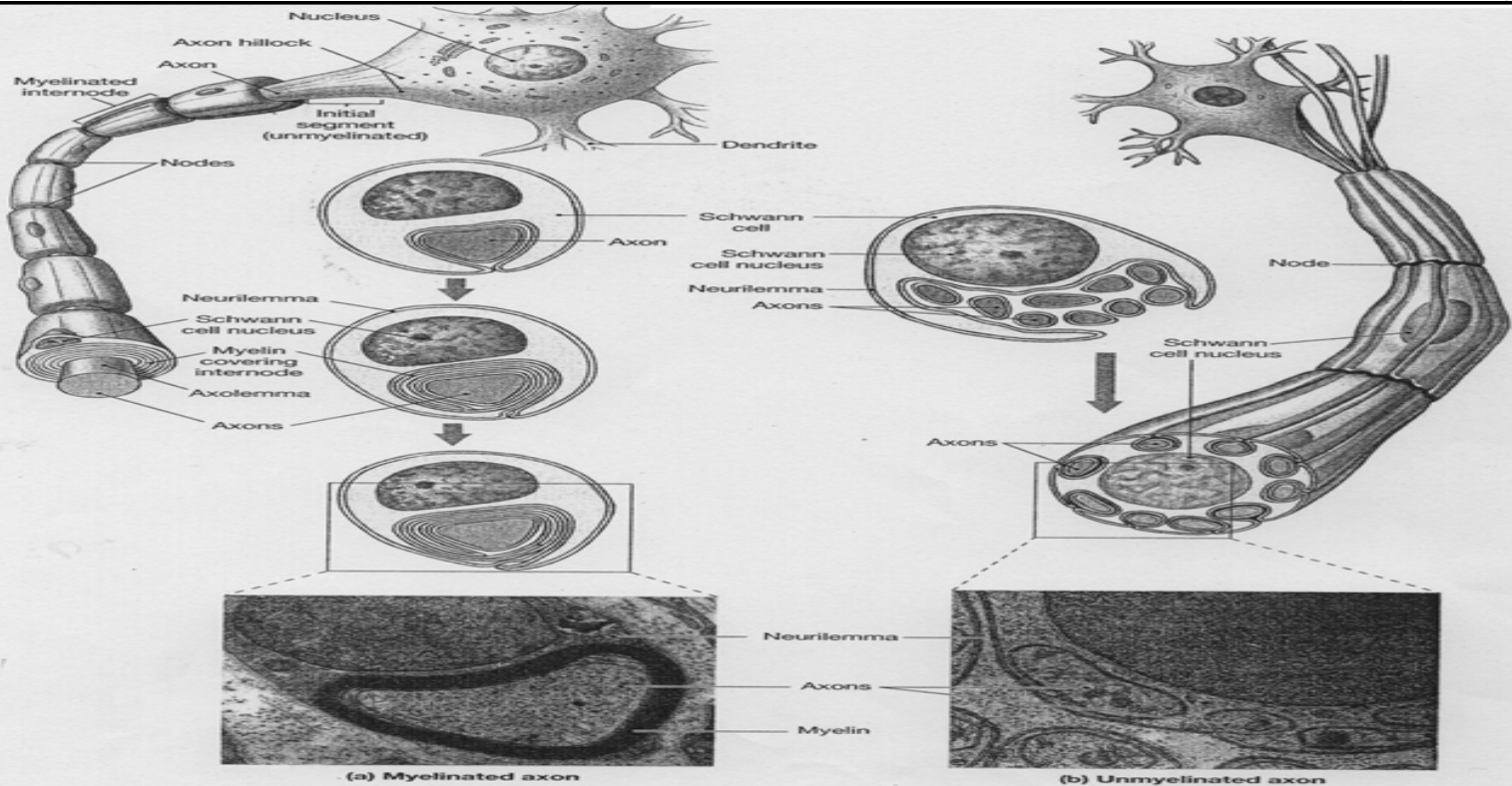
Nervous system

[?] Two types of the nerve fibers are found:

- I. **Myelinated Nerve Fibers:** with **sphingomyelin** myelin sheath acts as an excellent **insulator**
 - decreases the ion flow through the membrane about 5000- fold
 - It envelops the axon except at the nodes of Ranvier.
 - Myelin sheath loss is associated with delayed or blocked conduction.
- II. **Non-myelinated Nerve Fibers:** The axon is simply surrounded by Schwann cells without formation of myelin sheath.



Nervous system

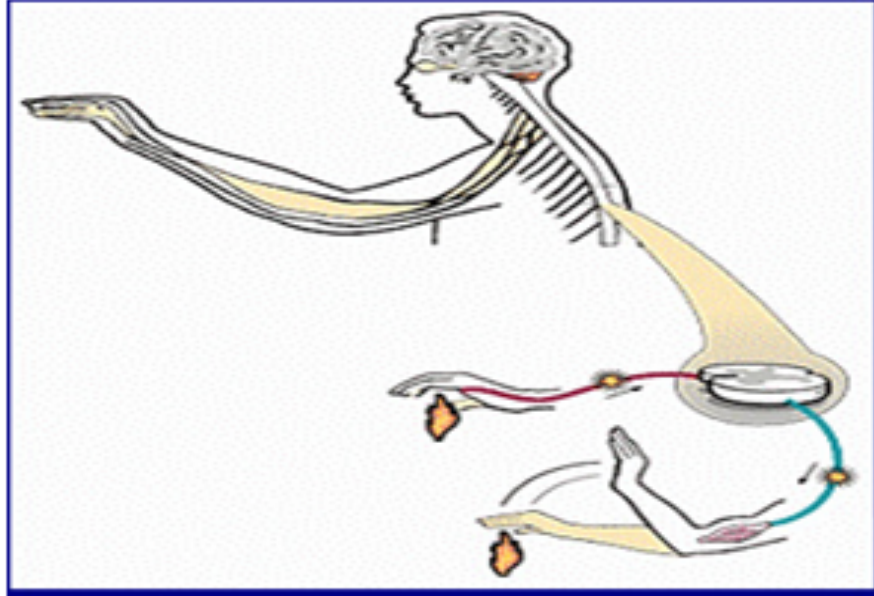


Nervous system

[?] The functional unit of the nervous system is the **reflex arc**: *homeostasis*

[?] Reflex arc is composed of **SRACEER**

- A proper **S**timulus.
- A sense organ (**R**eceptor).
- An **A**fferent neuron.
- One or more synapses (**C**enter).
- An **E**fferent neuron.
- An **E**ffector organ.
- Proper **R**esponse



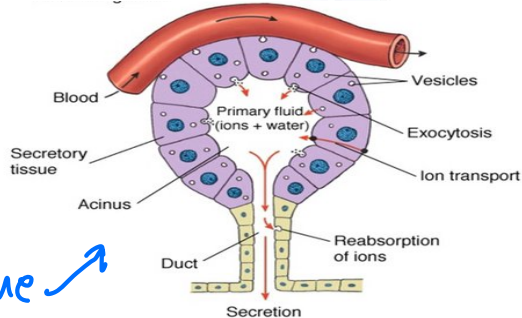
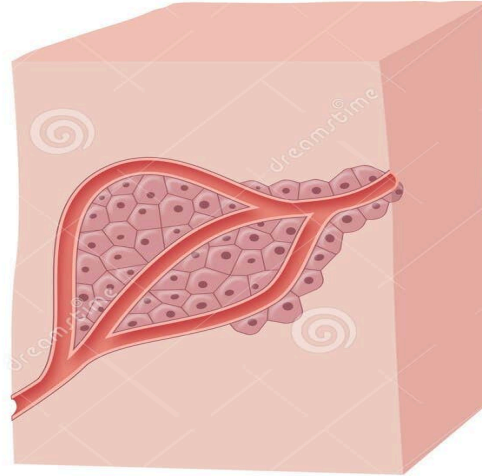
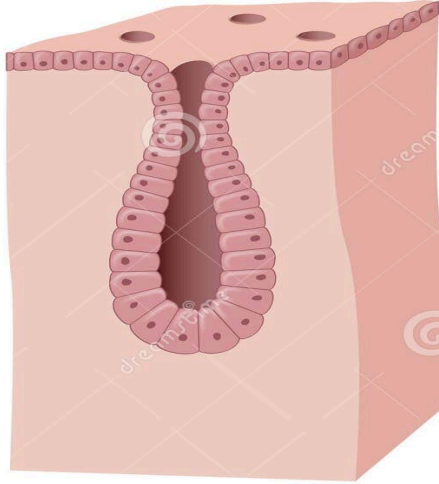
Regulatory systems

	<u>The endocrine system</u>	<u>The nervous system</u>
Structural unit		Neurons and neuroglia
Functional unit		Reflex arc
Onset		Rapid
Duration		Short
Example		Muscular contractions

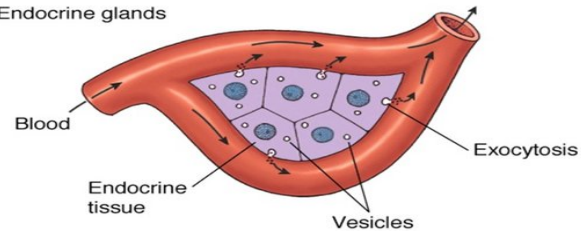


Endocrine glands.

Regulatory systems



Endocrine glands



exocrine ↗

Endocrine glands

[?] The endocrine system:

- Endocrine glands constitute a slow control system of the body.
- They produce their actions by secreting hormones.

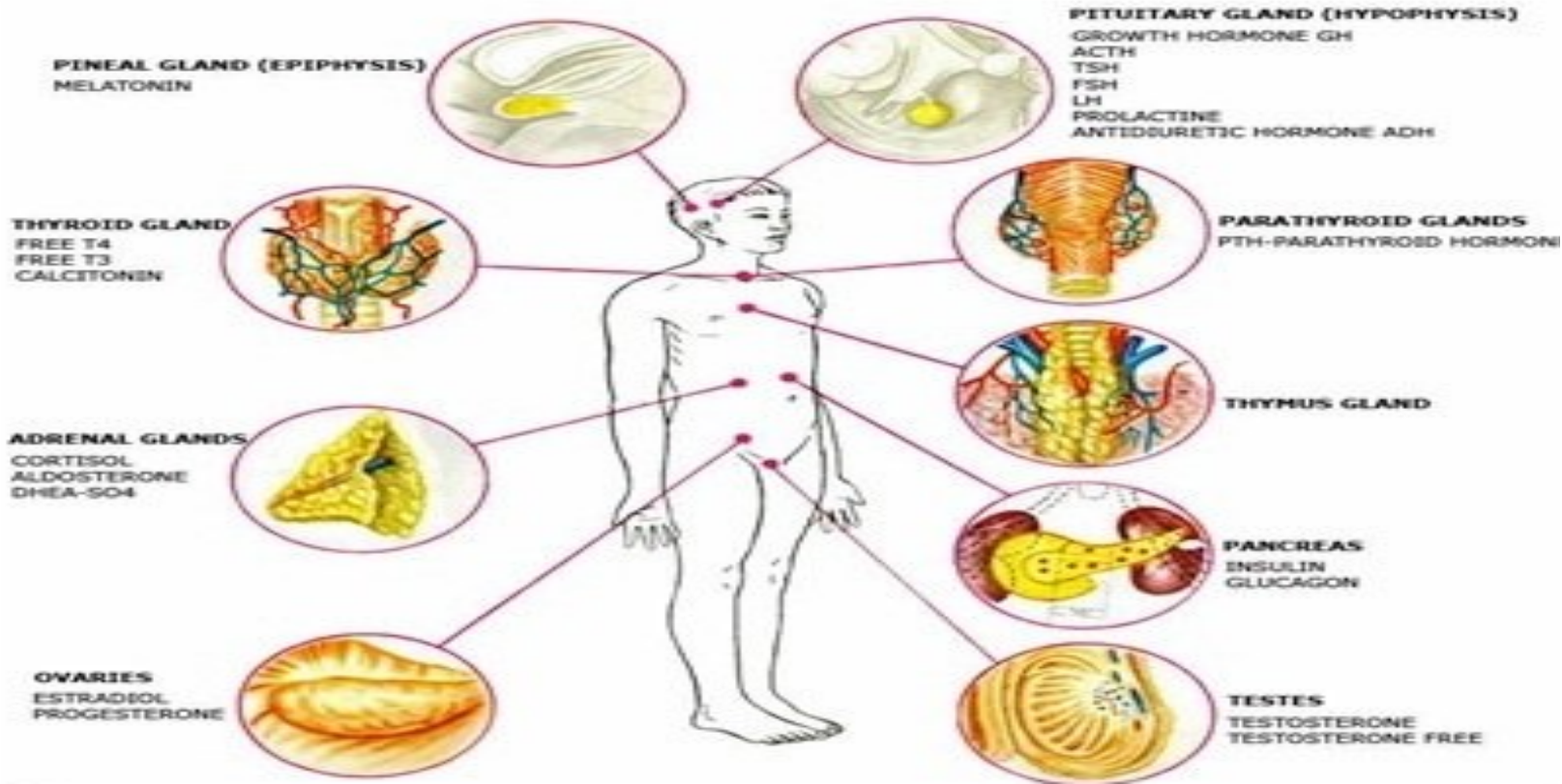
Co-ordination of Body Functions by Chemical Messengers:

1. Gap Junctions. (direct, fast)
2. Neurotransmitters (Synaptic): (rapid mechanism).
3. Endocrine hormones. (slow, systemic)
4. Paracrine. (local)
5. Autocrine.
6. Neuroendocrine hormones: (intermediate mechanism).

Regulatory systems

	<u>The endocrine system</u>	<u>The nervous system</u>
Structural unit	Ductless (endocrine glands)	Neurons and neuroglia
Functional unit	chemical messengers (hormones)	Reflex arc
Onset	Rapid, moderate or slow	Rapid
Duration	Short or Prolonged	Short
Example	Metabolic functions	Muscular contractions

Endocrine glands



Endocrine glands

- [?] Pituitary gland which is the master gland.
- [?] The pituitary is under the control of the Hypothalamus (neuro-endocrine).
- [?] Three endocrine glands under the control of pituitary gland:

1. Thyroid gland.
2. Suprarenal cortex.
3. Gonads (testis & ovaries)*.



*mixed
glands*

- [?] Three endocrine glands not under the control of pituitary gland:

1. Parathyroid glands.
2. Suprarenal medulla.
3. Pancreas*.



Endocrine glands

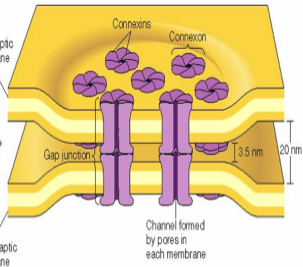
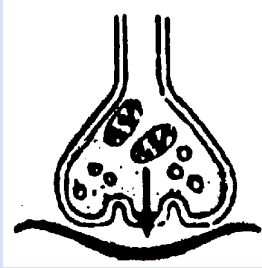
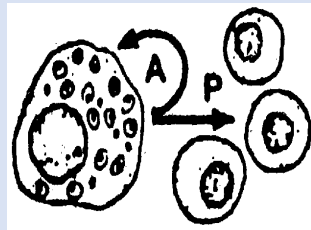
[?] In addition, there are organs which possess endocrine function together with their non-endocrine function:

- The heart secretes atrial natriuretic peptide (ANP).
- The kidney secretes erythropoietin, $1,25(\text{OH})_2$ cholecalciferol, renin among others.
- The liver secretes somatomedines, 25 hydroxy-cholecalciferol & hepcidin.
- The skin secretes calciferol (vitamin D).
- The gastrointestinal tract secretes GIT hormones as gastrin, CCK, secretin, VIP..... etc.
- The pineal gland secretes melatonin.



Intercellular communication

Intercellular communication

Gap Junctions	Synaptic	Paracrine & Autocrine	Endocrine
			
Directly from cell to cell	Across synaptic cleft	By diffusion in ISF	By circulating body fluids
Local	Local	Local	Systemic



Thank You!



Conclusion

- The endocrine and nervous systems are the 2 regulatory systems.
- The structural units of the nervous system are neurons and neuroglia, while the functional unit is the reflex arc.
- The endocrine system is one of the regulatory systems of the body.
- There are different methods of chemical transmission between cells
- Human body contains several endocrine glands.
- Both the endocrine and nervous systems are interconnected.

Case scenario

A child accidentally touches a hot stove and immediately withdraws her hand within a fraction of a second, with the response lasting only briefly. Which regulatory system is primarily responsible for this rapid, brief response, and why?

- 1) The endocrine system, because hormones act rapidly and briefly
- 2) The nervous system, because its onset and duration are both rapid and short
- 3) The endocrine system, because it directly controls skeletal muscle
- 4) Both systems equally, since they always act together with identical timing

References

Guyton and Hall, 13th edition. Unit II(6), Unit V(25); 305-14.

Ganong's review of medical physiology 25th ed. Section I(2, 5); 45-53.

Thank You

